



Draw a Square with a Loop

Ontario Math C3.1 — repeating events

Name: _____

Date: _____

★ I can use a for-loop to make Bit draw a square.

Four sides, one loop

A square has 4 equal sides and 4 square corners (right 90). Instead of typing forward and turn four times, a loop does it for you.

Read the code: `range(4)` repeats forward 50, right 90 — four times — and Bit draws a square.

Read this Python code

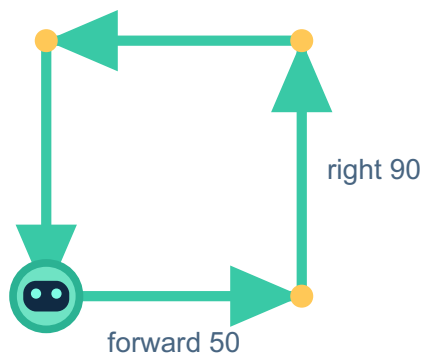
```
for i in range(4):  
    bit.forward(50)  
    bit.right(90)
```

PYTHON

OUTPUT

Bit draws a square and ends back at the start.

`range(4)` → a square



`range(4)` of forward 50, right 90 → a square.

1 Read it

How many sides does Bit draw? _____ How many right-90 turns? _____

2 Predict, then run

You change forward(50) to forward(100) (range(4) and right(90) stay). PREDICT: is it still a square? What changes — the shape or the size? Then run it.

3 Write your own

Finish the code so Bit draws a square with sides of 60. Fill in the two numbers: for i in range(____):
bit.forward(60) bit.right(____)

4 Challenge

Bit draws a square with forward(50), four times. How long is the whole path Bit draws all the way around? (This is the square's perimeter.)

Big idea

A square = repeat 4 of (forward, right 90). The forward number sets the SIZE; range(4) and right 90 keep it a SQUARE.